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Integrated image sensor and lens assembly

Abstract

An integrated image sensor and lens assembly is disclosed that includes an image sensor assembly defining an optical axis; a lens holder; and a lens barrel. The lens holder is coupled to the image sensor assembly and includes a tubular body portion terminating in a tapered surface that extends at an angle to the optical axis. The lens barrel is coupled to the lens holder by an adhesive that is applied between the lens holder and the lens barrel such that the adhesive extends in non-perpendicular relation to the optical axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/722,446, filed on Apr. 18, 2022, which is a continuation of U.S. patent application Ser. No. 16/735,183, filed on Jan. 6, 2020, now U.S. Pat. No. 11,336,807, which is a continuation of U.S. patent application Ser. No. 16/144,140, filed on Sep. 27, 2018, now U.S. Pat. No. 10,530,978, which is a continuation of U.S. patent application Ser. No. 14/705,891, filed on May 6, 2015, now U.S. Pat. No. 10,122,902, which claims the benefit of U.S. Provisional Patent Application Ser. No. 62/019,146, filed Jun. 30, 2014, and U.S. Provisional Patent Application Ser. No. 61/990,053, filed May 7, 2014. The subject matter of each of the above-identified applications is incorporated herein by reference in its entirety.

BACKGROUND

(1) Manufacturing of lens assemblies for high-resolution cameras typically require a high degree of precision in positioning components of the lens assembly to ensure that the lens will achieve proper focus. As a result, a challenge exists in achieving a fast, automated, and high-yielding assembly process for high-resolution cameras.

(2) In a conventional manufacturing process, a lens barrel housing the camera lens is placed within a housing assembly affixed to an image sensor. Upon testing the lens barrel to position it for proper alignment, the lens barrel is affixed to the housing assembly using a “floating” assembly process whereby the lens barrel is attached to the housing using adhesive between a ridge of the lens barrel extending parallel to the image plane and an inner lip of the housing extending parallel to the image plane. The adhesive may expand and/or contract when cured, or after curing, the adhesive may expand and/or contract over time based on changes in temperature or other conditions. Because the adhesive is placed between surfaces of the lens barrel and housing that are parallel to the image plane, the expansion or contraction of the adhesive causes the lens to shift along the optical axis, thus altering the distance between the lens and the image sensor. This affects focus of the lenses and compromises performance and yield.

Description

DETAILED DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an exploded view of an example integrated image sensor and lens assembly, according to one embodiment.
- (2) FIG. 2 illustrates a cross-sectional view of the lens holder and lens barrel sliced along a plane parallel to the optical axis, according to one embodiment.
- (3) FIG. 3 illustrates a cross-sectional view of the lens holder and lens barrel sliced along a plane parallel to the optical axis, according to one embodiment.
- (4) FIG. 4 illustrates a cross-sectional view of the lens holder and lens barrel sliced along a plane parallel to the optical axis, according to one embodiment.
- (5) FIG. 5 illustrates an example integrated sensor and lens assembly undergoing a curing process during manufacture, according to an embodiment.
- (6) FIG. 6 is an enlarged view of the example integrated sensor and lens assembly undergoing the curing process during manufacture as illustrated in FIG. 5.
- (7) FIG. 7 illustrates an example integrated sensor and lens assembly in an example camera body, according to one embodiment.
- (8) FIG. 8 illustrates an example camera that includes the example integrated image sensor and lens assembly of FIG. 1.

DETAILED DESCRIPTION

(9) The figures and the following description relate to preferred embodiments by way of illustration only. It should be noted that from the following discussion, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of what is claimed.

(10) Reference will now be made in detail to several embodiments, examples of which are illustrated in the accompanying figures. It is noted that wherever practicable similar or like reference numbers may be used in the figures and may indicate similar or like functionality. The figures depict embodiments of the disclosed system (or method) for purposes of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

(11) Configuration Overview

(12) In an embodiment, an integrated image sensor and lens assembly comprises a lens barrel adhered to a lens holder using an adhesive ring applied between an exterior surface of the lens barrel and an interior surface of the lens holder that are oriented in a direction perpendicular to the image plane (parallel to the optical axis). The lens barrel and lens holder are bonded based on the sheer strength (rather than the tensile strength) of the adhesive. Thus, adhesive expansion does not substantially affect the distance between the lens and the image sensor and therefore does not affect the focal plane of the assembly. As described herein, substantially constant refers to the optical distance is within a predefined tolerance that would be acceptable to those skilled in art. In various embodiments, the predefined tolerance can be, for example, a 1% tolerance, a 2% tolerance, a 5% tolerance, etc. While expansion or contraction of the adhesive may still occur, the forces are directed radially (rather than axially) and are counteracted because the adhesive is applied concentrically.

(13) In another embodiment, an integrated image sensor and lens assembly comprises a lens barrel adhered to an image sensor assembly using an adhesive applied between a surface of the lens barrel and an outer side wall of the image sensor assembly that are oriented in a direction perpendicular to the image plane (parallel to the optical axis). The lens barrel and image sensor assembly are bonded based on the sheer strength (rather than the tensile strength) of the adhesive. Thus, adhesive expansion does not substantially affect the distance between the lens and the image sensor and therefore does not affect the focal plane of the assembly. While expansion or contraction of the

adhesive may still occur, the forces are directed radially (rather than axially) and are counteracted because the adhesive is applied concentrically.

(14) FIG. 1 illustrates an exploded view of an embodiment of an integrated image sensor and lens assembly **100**. The integrated image sensor and lens assembly **100** comprises an image sensor substrate **140**, an image sensor assembly **130**, a lens holder **120**, and a lens barrel **110**. The image sensor substrate **140** comprises a printed circuit board for mounting the image sensor assembly **130** and may furthermore include various electronic components that operate with the image sensor assembly **130** or provide external connections to other components of the camera system. The image sensor assembly **130** houses an image sensor (e.g., a high-definition image sensor) for capturing images and/or video and includes structural elements for physically coupling the image sensor assembly **130** to the image sensor substrate **140** and to the lens holder **120**. The image sensor of the image sensor assembly **130** lies on an image plane **160**. The lens holder **120** physically couples with the image sensor assembly **130** and the lens barrel **110**. In one embodiment, the lens holder **120** comprises a base portion **124** and a tube portion **122**. The base portion **124** includes a bottom surface in a plane substantially parallel to a surface of the image sensor substrate **140**. Furthermore, the base portion **124** includes a recess (not viewable in FIG. 1) to enable the bottom surface of the base portion **124** to lie flat against the image sensor substrate **140** while partially enclosing the image sensor assembly **130**. The tube portion **122** of the lens holder **120** extends away from the image sensor assembly **130** along the optical axis **150** and includes a substantially cylindrical channel for receiving the lens barrel **110**. The lens barrel **110** comprises one or more lenses or other optical components to direct light to the image sensor assembly **130**. The lens barrel **110** comprises a lower portion **116**, one or more barrel arms **114**, and a lens window **112**. The lower portion of the lens barrel **116** is substantially cylindrical and structured to at least partially extend into the channel of the tube portion **122** of the lens holder **120**. The lens arms **114** extend radially from the body of the lens barrel **110** and are outside the channel of the lens holder **120** when assembled. The lens arms **114** may be used to physically couple the lens barrel **110** to the camera body (not shown). The lens window **112** includes optical components to enable external light to enter the lens barrel **110** and be directed to the image sensor assembly **130**,

(15) FIG. 2 illustrates a cross-sectional view of the lens holder **120** and lens barrel **110** sliced along a plane parallel to the optical axis **150**. Interior components (e.g., lenses) of the lens barrel **110** are omitted to more clearly show the structural features of the lens barrel **110**. As illustrated in FIG. 2, the lower portion **116** of the lens barrel **110** partially extends into the tube portion **122** of the lens holder **120**. An adhesive **234** is applied between an interior surface of the tube portion **122** that is oriented parallel to the optical axis **150** and an exterior surface of the lower portion **116** of the lens barrel **110** (also oriented parallel to the optical axis **150**) to radially bond the lens barrel **110** to the lens holder **120**. The section of the lower portion **116** of the lens barrel **110** that extends into the tube portion **122** of the lens holder **120** is adhered to the tube portion **122** via the adhesive **234**. The lens barrel **110** and lens holder **120** are bonded based on the shear strength of the adhesive **234**. If expansion of the adhesive **234** occurs, the forces will be directed radially (i.e., along a plane parallel to the image plane) and will therefore not substantially affect the distance between the lens of the lens barrel **110** and the image plane **160**.

(16) In the illustrated embodiment, the lower portion **116** of the lens barrel **110** includes a plurality of ridges **236** on the exterior surface. The ridges **236** increase the surface area on the exterior surface of the lower portion **116** for the adhesive **234** to bond to and thereby increase the adhesive strength.

(17) The illustrated embodiment also shows a gasket **232** positioned at a lower end of the lens barrel **110** just below the adhesive **234**. The gasket **232** may be used in the manufacturing process as a barrier to prevent the adhesive **234** from overflowing and potentially reaching the image sensor if the adhesive **234** is insufficiently viscous. In one embodiment, the gasket **232** comprises a pliable

material (e.g., rubber, plastic, etc.) so that it does not prevent fine adjustments from being made to the positioning of the lens barrel **110** relative to the image sensor assembly **130** along the optical axis during the assembly and alignment process. The pliability of the gasket **232** furthermore absorbs any forces along the optical axis caused by expansion or contraction of the adhesive **234**.

(18) FIG. 3 illustrates a cross-sectional view of the lens holder **120** and lens barrel **110** sliced along a plane parallel to the optical axis **150**. In the illustrated example, the interior surface of the lens barrel **110** is substantially smooth and the adhesive **234** is applied to the interior surface of the lens barrel **110**,

(19) FIG. 4 illustrates a cross-sectional view of the lens holder **120** and lens barrel **110** sliced along a plane parallel to the optical axis **150**. The gasket **232** is omitted in this illustrated embodiment.

(20) FIG. 5 illustrates an embodiment of an integrated sensor and lens assembly undergoing a curing process during manufacture. In the illustrated embodiment, the lens holder **110** includes one or more ports **506** that enable ultraviolet (UV) rays to pass through from the LED UV system **504**. The UV rays cure the adhesive **234** and thereby to lock the lens barrel **110** and lens holder **120** in place after tested and properly aligned. A top ring UV cure **502** is also applied to the top of the lens holder **110**. In one embodiment, the lens holder **120** includes four ports **506** equally spaced around the circumference of the tube portion **122** of the lens holder **110**. Alternatively, three equally spaced ports **506** or a different number or configuration of ports **506** may be used.

(21) The curing process settles the lens barrel **110** in the lens holder to provide the desired modulation transfer function (MTF) and focuses position of the lenses prior to further thermal curing. As illustrated in FIG. 6, the adhesive has a curing depth such that it immobilizes the settled lens barrel **110** prior to thermal curing. Particularly, the depth of the adhesive from the external opening of the pot **506** to the interior surface of the lens barrel **110** (for example, the depth **610** in FIG. 6) locks the lens barrel **110** in place even before curing is applied.

(22) FIG. 7 illustrates another alternative embodiment of an integrated sensor and lens assembly **702** for use in a camera body **700**. The lens assembly **702** comprises a lens barrel **110**, an image sensor substrate **140**, and an image sensor assembly **130**. Similar to the embodiment illustrated in FIG. 1, the image sensor substrate **140** comprises a printed circuit board for mounting the image sensor assembly **130** and may furthermore include various electronic components that operate with the image sensor assembly **130** or provide external connections to other components of the camera system. The image sensor assembly **130** houses an image sensor (e.g., a high-definition image sensor) for capturing images and/or video and includes structural elements for physically coupling the image sensor assembly **130** to the image sensor substrate **140** and to lens barrel **110**. The image sensor of the image sensor assembly **130** lies on an image plane **160**.

(23) The lens barrel **110** comprises one or more lenses or other optical components to direct light to the image sensor assembly **130**. The lens barrel **110** comprises a lower portion **116** and an upper portion **704** having a lens window **112**. The lower portion of the lens barrel **116** includes a channel **706** for receiving the image sensor assembly **130**. The channel **706** can be substantially cylindrical, square, rectangular, or other shaped. The lens barrel **110** may be physically coupled to the camera body **700**. The lens window **112** includes optical components to enable external light to enter the lens barrel **110** and be directed to the image sensor assembly **130**. The lens barrel **110** is positioned such that the image sensor assembly **130** protrudes into the channel **706**. An interior surface of the lens barrel **110** is oriented to be substantially perpendicular to the image plane **160** and substantially parallel to the optical axis **150**.

(24) The image sensor assembly **130** is adhered to the interior surface of the lens barrel **110** directly via the adhesive **234**, without a separate lens holder. The image sensor assembly **130** includes a plurality of side walls oriented in a direction substantially perpendicular to the substrate **140**. The adhesive **234** is applied between the interior surface of the lens barrel **110** and an outer side wall of the image sensor **130**. The integrated sensor and lens assembly **702** is substantially cylindrical, square, or rectangular.

(25) In one embodiment, the channel **706** of the lens barrel **110** is substantially square or rectangular and the image sensor assembly **130** is substantially square or rectangular. The channel **706** of the lens barrel **110** mates with the image sensor assembly **130** such that an interior surface of the channel **706** of the lens assembly **1302** adheres to the outer side wall of the image sensor **130**. In another embodiment, the channel **706** of the lens barrel **110** may be substantially circular and mate with a substantially circular image sensor **130**. In yet another embodiment (not shown), an adhesive **234** is applied between a top surface of the image sensor assembly **130** and a shelf that protrudes from an interior surface of the lens barrel **112** in a direction substantially parallel to the top surface of the image sensor **130**. Other embodiments and features described above (e.g., the use of gasket **232**, the use of ridges **236**) may be combined with the embodiment of FIG. **13** in which the lens assembly **702** includes a lens barrel adhered to the image sensor assembly **130** via the adhesive **234** directly. The gasket may be positioned at a lower end of the lens barrel **110** just below the adhesive **234**.

(26) Example Camera System Configuration

(27) FIG. **8** illustrates an embodiment of an example camera **800** that includes the integrated image sensor and lens assembly **100** described above. The camera **800** comprises a camera body having a camera lens structured on a front surface of the camera body, various indicators on the front of the surface of the camera body (such as LEDs, displays, and the like), various input mechanisms (such as buttons, switches, and touch-screen mechanisms), and electronics (e.g., imaging electronics, power electronics, etc.) internal to the camera body for capturing images via the camera lens and/or performing other functions. The camera **800** is configured to capture images and video, and to store captured images and video for subsequent display or playback. As illustrated, the camera **800** includes a lens **802** configured to receive light incident upon the lens and to direct received light onto an image sensor internal to the lens. The lens **802** is enclosed by a lens ring **804**, which are both part of the integrated image sensor and lens assembly **100** discussed above.

(28) The camera **800** can include various indicators, including the LED lights **806** and the LED display **808**. The camera **800** can also include buttons **810** configured to allow a user of the camera to interact with the camera, to turn the camera on, and to otherwise configure the operating mode of the camera. The camera **800** can also include a microphone **812** configured to receive and record audio signals in conjunction with recording video. The side of the camera **800** includes an I/O interface **814**.

(29) Additional Configuration Considerations

(30) Throughout this specification, some embodiments have used the expression “coupled” along with its derivatives. The term “coupled” as used herein is not necessarily limited to two or more elements being in direct physical or electrical contact. Rather, the term “coupled” may also encompass two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other, or are structured to provide a thermal conduction path between the elements.

(31) Likewise, as used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(32) In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

(33) Finally, as used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in

various places in the specification are not necessarily all referring to the same embodiment. (34) Upon reading this disclosure, those of skill in the art will appreciate still additional alternative structural and functional designs for a camera expansion module as disclosed from the principles herein. Thus, while particular embodiments and applications have been illustrated and described, it is to be understood that the disclosed embodiments are not limited to the precise construction and components disclosed herein. Various modifications, changes and variations, which will be apparent to those skilled in the art, may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope defined in the appended claims.

Claims

1. A method of assembling an image sensor and lens assembly for an image capturing device, the method comprising: coupling a first end of a lens holder to an image sensor assembly such that an angled, chamfered rim at a second end of the lens holder extends at an angle to an optical axis defined by the image sensor assembly, wherein the angled, chamfered rim includes a tapered surface, wherein the tapered surface extends continuously between first and second ends thereof; inserting a single-piece lens barrel into the lens holder; and coupling the single-piece lens barrel to the lens holder by applying an adhesive between the lens holder and the single-piece lens barrel such that the adhesive extends in non-perpendicular relation to the optical axis.
2. The method of claim 1, wherein inserting the single-piece lens barrel into the lens holder includes inserting ridges extending partially along an overall length of the single-piece lens barrel into the lens holder.
3. The method of claim 1, wherein inserting the single-piece lens barrel into the lens holder includes inserting ridges extending non-continuously between first and second ends of the single-piece lens barrel into the lens holder.
4. The method of claim 1, wherein inserting the single-piece lens barrel into the lens holder includes inserting the single-piece lens barrel into a cylindrical channel defined by the lens holder.
5. The method of claim 1, wherein inserting the single-piece lens barrel into the lens holder includes inserting a first end of the single-piece lens barrel into the lens holder such that a second end of the single-piece lens barrel is exposed therefrom, and wherein the single-piece lens barrel tapers inwardly from the second end towards the first end.
6. The method of claim 1, further comprising: positioning one or more lenses within the single-piece lens barrel to direct light to the image sensor assembly.
7. The method of claim 1, further comprising: enclosing the image sensor assembly within the lens holder.
8. The method of claim 7, wherein enclosing the image sensor assembly within the lens holder includes inserting the image sensor assembly into a base portion of the lens holder positioned at the first end thereof, and wherein the base portion of the lens holder extends radially outward in relation to a body portion of the lens holder.
9. The method of claim 8, wherein inserting the image sensor assembly into the base portion includes inserting the image sensor assembly into a recess defined by the base portion.
10. The method of claim 7, wherein enclosing the image sensor assembly within the lens holder includes inserting the image sensor assembly into a base portion of the lens holder extending radially outward in relation to a body portion thereof, wherein the body portion defines a first transverse cross-sectional configuration, and wherein the base portion defines a second transverse cross-sectional configuration that is different than the first transverse cross-sectional configuration.
11. The method of claim 7, wherein enclosing the image sensor assembly within the lens holder includes inserting the image sensor assembly into a base portion of the lens holder extending radially outward in relation to a body portion thereof, wherein the body portion defines an annular

transverse cross-sectional configuration, and wherein the base portion defines a non-annular transverse cross-sectional configuration.

12. The method of claim 1, further comprising: orienting the single-piece lens barrel such that a radial extension on the single-piece lens barrel is positioned externally of the lens holder.

13. The method of claim 1, further comprising: orienting the single-piece lens barrel such that a radial extension on the single-piece lens barrel is positioned externally of the lens holder, wherein the radial extension is located between ridges extending along an exterior surface of the single-piece lens barrel and an end of the single-piece lens barrel.

14. A method of assembling an image sensor and lens assembly for an image capturing device, the method comprising: providing: an image sensor assembly defining an optical axis; a lens holder including: an angled, chamfered rim at an outer end of the lens holder extending at an angle to the optical axis, wherein the angled, chamfered rim defines an uppermost surface and a tapered surface devoid of threading; and an inner surface extending from the angled, chamfered rim such that the tapered surface extends between the uppermost surface and the inner surface; and a lens barrel having a first end and a second end; coupling the lens holder to the image sensor assembly; and adhesively securing the lens barrel within the lens holder.

15. The method of claim 14, wherein adhesively securing the lens barrel within the lens holder includes inserting the lens barrel into the lens holder such that external ridges extending non-continuously between the first end of the lens barrel and the second end of the lens barrel are positioned within a cylindrical channel defined by the lens holder.

16. The method of claim 15, further comprising: orienting the lens barrel such that a radial extension on the lens barrel is located between the external ridges and the second end of the lens barrel, whereby the radial extension is spaced from the lens holder along the optical axis.

17. The method of claim 14, wherein coupling the lens holder to the image sensor assembly includes inserting the image sensor assembly into a base portion of the lens holder extending radially outward in relation to a body portion thereof, wherein the body portion defines an annular transverse cross-sectional configuration, and wherein the base portion defines a non-annular transverse cross-sectional configuration.

18. A method of assembling an image sensor and lens assembly for an image capturing device, the method comprising: coupling a lens holder to an image sensor assembly defining an optical axis; inserting a lens barrel into the lens holder; and securing the lens barrel to the lens holder by applying an adhesive to an angled, chamfered rim terminating at an end of the lens holder and extending in non-perpendicular relation to the optical axis.

19. The method of claim 18, wherein inserting the lens barrel into the lens holder includes inserting the lens barrel into a body portion of the lens holder including: an inner surface; and a non-threaded surface defined by the angled, chamfered rim and extending: radially outward from the inner surface; and axially along the optical axis, wherein the non-threaded surface includes first and second ends and extends continuously therebetween.

20. The method of claim 18, further comprising: orienting the lens barrel such that a radial extension on the lens barrel located between first and second ends thereof is spaced from the lens holder along the optical axis.
